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Signals for Next Season

2027 batting forecasts and scouting investigation policies using 2026 metrics

Full 2026 regular season · Hitters · Annual forecasting study; early detection is not evaluated
Original forecasts frozen: 2026-10-02T21:30:02.338743+00:00 · 2027 evaluation rules frozen: 2026-10-02T23:49:10.564896+00:00

Explanatory and diagnostic revision: 2026-10-04 UTC · Models, 298 forecasts, and evaluation rules unchanged

Conclusion: Additional signals have not established a superior selection policy

The primary analysis forecasts next-season wOBA (a batting metric weighted by run contribution). Current skill levels, including xwOBA (an expected metric reflecting contact quality), improved observed retention AUC (ranking discrimination between successes and failures). However, the 2026 M3−M1 AUC difference was +.096 [.009, .181] conditional on PA and +.072 [−.002, .155] across all followed hitters. The latter interval includes zero and does not establish an advantage from change variables. RMSE measures prediction error; lower is better.

Growth success requires at least 200PA (plate appearances) next season and a wOBA increase of at least +.020. This outcome does not separate skill development from rebound through regression to the mean. The budget counts player-season investigation slots. In 2022–2026, the original signal screen achieved 7/53 growth successes versus M1's 13/53; the central 95% random-selection range was 6–17. The revised rule achieved 2/22 versus 7/22, with a random range of 2–9. Neither method was clearly distinguishable from this random growth benchmark.

Keeping M1 as the default forecast is an operating decision in the absence of established screen superiority, not proof that M1 is superior. Historical comparisons are exploratory, and 2026 informed selection. The 2027 list of at most 5 hitters is a case review; adoption and stopping decisions follow a separately frozen multi-season plan. The original 2027 forecast and evaluation freeze is preserved.

Python code, executed notebook, and reproducibility files · Glossary · α sensitivity · Original forecast and evaluation dates and SHA256 hashes are preserved.

One-page application summary · Decision follow-up rules

Question Validation Forecasts & hypotheses Player evidence Decision Scope & reproducibility Python code Glossary

01 · DECISION & ANALYTICAL QUESTION

Which signals should inform next-season scouting?

> Do current underlying skills and year-to-year changes improve next-season batting forecasts and breakout-retention assessment beyond traditional results?

The operating decision is how to allocate limited video-review time across hitters. The original early-detection question exceeded what annual data could test, so it was removed from the main title and question. This report evaluates next-season forecasts and investigation policies after the season ends.

The primary analysis predicts actual next-season wOBA. Breakout retention remains **the previously specified secondary question**; stronger results do not promote it to the primary outcome. The policy comparison was added as an exploratory response to critique. This study does not evaluate total acquisition value, including defense, baserunning, playing time, or contract price.

02 · DATA & DESIGN

Validate on history; freeze the 2027 forecasts

2015–2026	1,806	298	2027 pending
Source seasons	Final training player-season pairs	Eligible hitters with 2026 inputs	Future outcomes unobserved

Eligibility requires at least 200PA in the input season and 100PA in the preceding season. The target is actual next-season wOBA among hitters with at least 200PA. Hitters with insufficient playing time or no subsequent MLB record remain unsuccessful outcomes in scouting-policy evaluation.

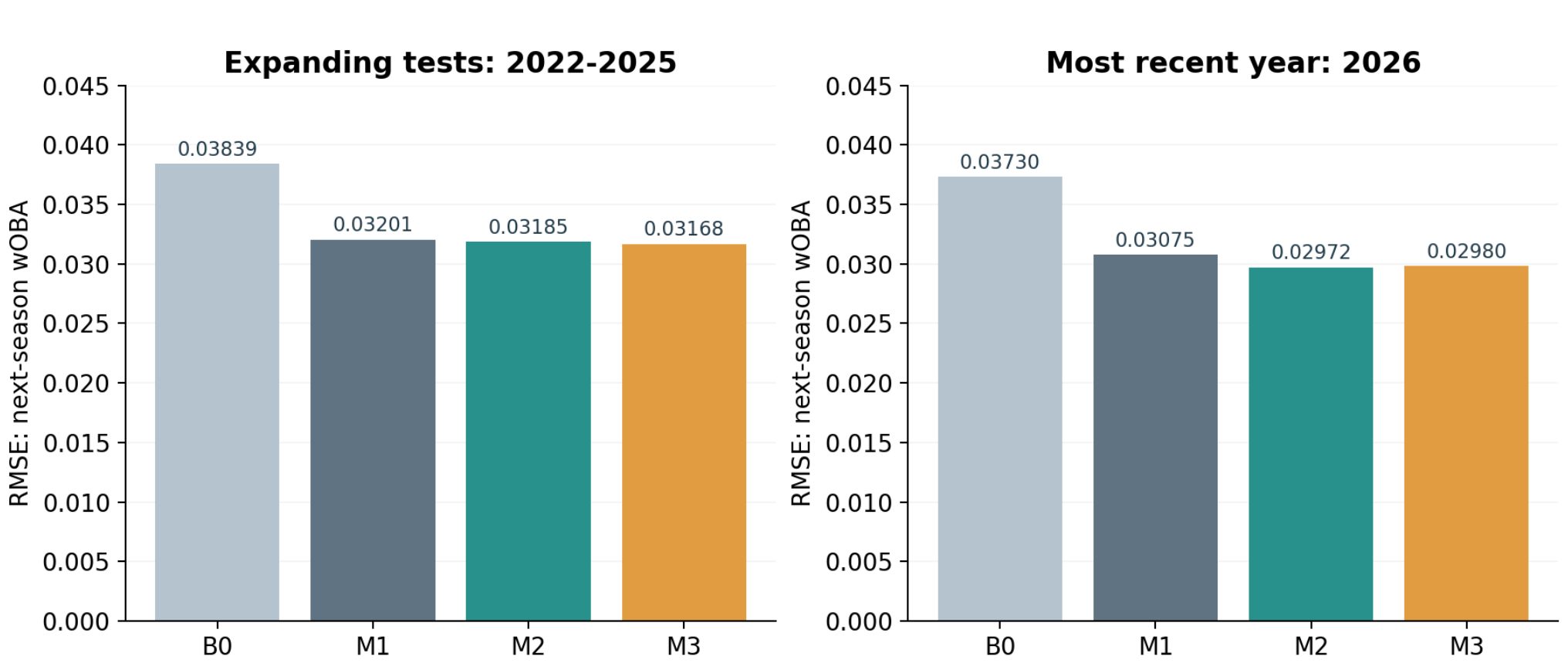
For 2022–2025, each target season is evaluated separately using **only outcomes completed before that target year**. The 2026 evaluation uses 2025 inputs. Completed outcomes through 2026 are then used to train forecasts from 2026 inputs for 2027.

M1: current/prior wOBA, age/age^2, current/prior PA, K%, BB%, and ISO. M2: M1 + xwOBA, barrel%, hard-hit%, EV, chase%, and whiff%. M3: M2 + year-to-year changes in those 6 metrics. Ridge α=10 is fixed; imputation and scaling are fitted within each training sample.

03 · PRIMARY VALIDATION

Overall batting forecasts: Change variables add little

Period	Model	RMSE ↓	MAE ↓	PA-weighted RMSE ↓
2022–2025 · 889 observations	B0 · Carry forward	0.03839	0.03031	0.03698
2022–2025 · 889 observations	M1 · Traditional	0.03201	0.02537	0.03058
2022–2025 · 889 observations	M2 · + Skill levels	0.03185	0.02538	0.03023
2022–2025 · 889 observations	M3 · + Skill changes	0.03168	0.02521	0.03010
2026 · 229 observations	B0 · Carry forward	0.03730	0.02961	0.03635
2026 · 229 observations	M1 · Traditional	0.03075	0.02422	0.03004
2026 · 229 observations	M2 · + Skill levels	0.02972	0.02393	0.02890
2026 · 229 observations	M3 · + Skill changes	0.02980	0.02399	0.02907



M3−M2 improved development RMSE by approximately 0.00017 wOBA, but worsened 2026 RMSE by approximately 0.00008. Development mean squared error improved by 1.10e-05; the exploratory year-cluster 95% interval was [-5.58e-06, 2.64e-05], including zero. The player-cluster

interval also included zero.

The conservative adoption rule required M3 to improve pooled RMSE and MAE, improve in at least 3 of 4 development years, and improve 2026 RMSE. It did not pass. M2 also failed because its development MAE was marginally higher than M1's. **The official 2027 ranking retains M1.** This follows the specified operating rule; it does not prove M1 is best in every setting.

The 2026 comparison also informed model selection, so it is not a second independent validation of the selected policy. Numerical precision supports computational checks, not equally precise baseball forecasts.

Additional diagnostic: Sensitivity to Ridge α

α=0.1/1/10/100/1000 were compared using identical time splits, variables, and training-only preprocessing. The official α=10 was retained; no optimum was selected from 2026 results. This additional diagnostic is exploratory because 2026 outcomes had already been observed.

α	Period	M1 RMSE	M2 RMSE	M3 RMSE
0.1	2022–2025 · 889 observations	0.03198	0.03184	0.03163
0.1	2026 · 229 observations	0.03071	0.02968	0.02977
1.0	2022–2025 · 889 observations	0.03199	0.03185	0.03164
1.0	2026 · 229 observations	0.03071	0.02969	0.02978
10.0	2022–2025 · 889 observations	0.03201	0.03185	0.03168
10.0	2026 · 229 observations	0.03075	0.02972	0.02980
100.0	2022–2025 · 889 observations	0.03198	0.03175	0.03169
100.0	2026 · 229 observations	0.03070	0.02972	0.02979
1000.0	2022–2025 · 889 observations	0.03251	0.03187	0.03186
1000.0	2026 · 229 observations	0.03034	0.02959	0.02963

At all five α values, M3's 2026 RMSE exceeded M2's, supporting the interpretation that change variables do not add consistent gains. However, M2's development MAE was marginally worse than M1's at α=10 and lower at α=100 and 1000. M1 adoption depends on the chosen α and operating rule, not proof of universal superiority. Diagnostic predictions at α=10 were checked against the saved originals.

RMSE, MAE, and conditional retention AUC at every α: supplement/ridge_alpha_sensitivity.csv. The next section separately reports the existing 36 retention-threshold combinations (.010/.020/.030 × .000/.010/.020 × PA 100/200/300/400).

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04 · SECONDARY QUESTION, WITH ROBUSTNESS

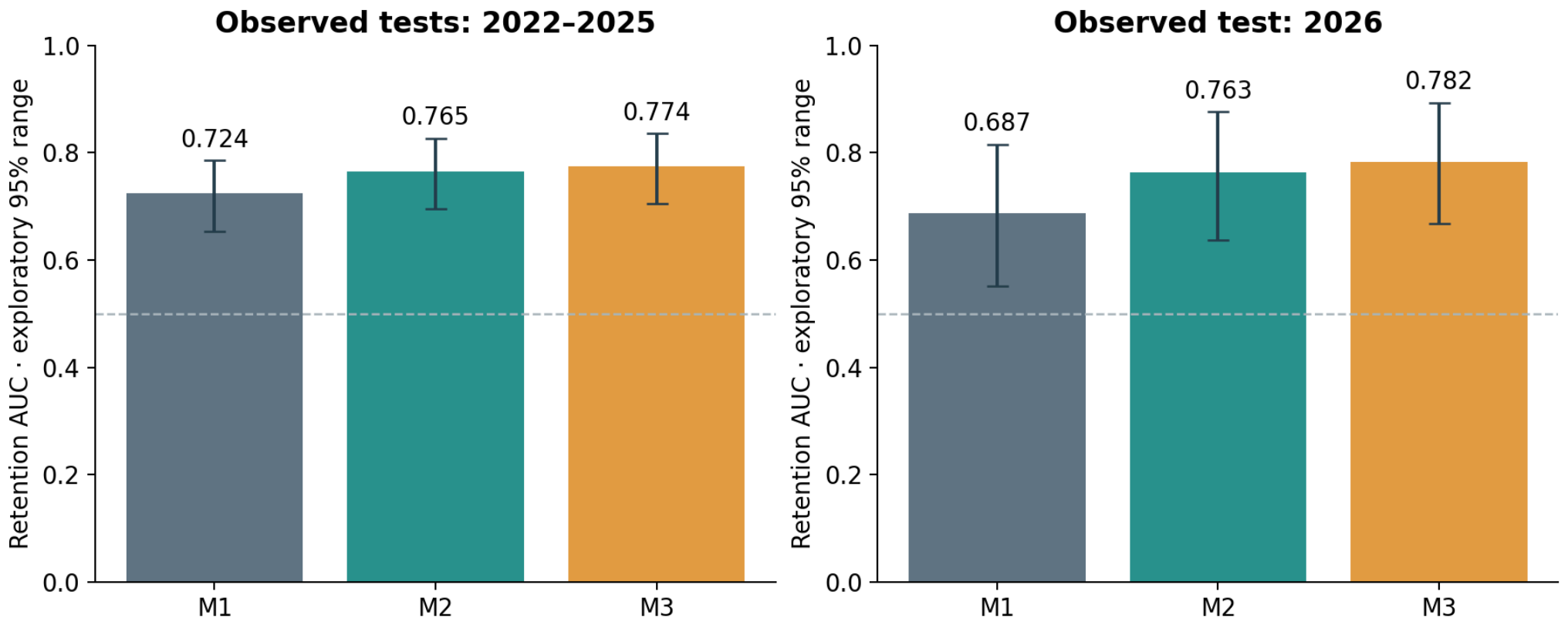
Retention discrimination: Samples, paired comparisons, and definitions

The main definition requires a wOBA rise ≥.020, next-season loss ≤.010, and next-season PA ≥200. This is an operating definition, not a unique or universal baseball standard.

Repeated observations: The development conditional analysis has 250 records from 206 hitters. All-followed breakout tracking has 292 records from 243 hitters; 42 did not reach 200PA next season. In 2026, only 66 of 86 hitters reached 200PA, excluding 20 from the conditional analysis. Both populations are disclosed.

Period	Population	Model	Records	Hitters	Retention successes	AUC	Exploratory 95% interval
2022–2025	Conditional on sufficient PA	M1	250	206	85	0.724	0.654–0.786
2022–2025	Conditional on sufficient PA	M2	250	206	85	0.765	0.695–0.827
2022–2025	Conditional on sufficient PA	M3	250	206	85	0.774	0.705–0.836
2022–2025	All followed: PA + retention	M1	292	243	85	0.694	0.624–0.760
2022–2025	All followed: PA + retention	M2	292	243	85	0.732	0.664–0.798

2022–2025	All followed: PA + retention	M3	292	243	85	0.743	0.674–0.811
2026	Conditional on sufficient PA	M1	66	66	25	0.687	0.552–0.815
2026	Conditional on sufficient PA	M2	66	66	25	0.763	0.637–0.876
2026	Conditional on sufficient PA	M3	66	66	25	0.782	0.668–0.893
2026	All followed: PA + retention	M1	86	86	25	0.648	0.518–0.773
2026	All followed: PA + retention	M2	86	86	25	0.711	0.582–0.822
2026	All followed: PA + retention	M3	86	86	25	0.720	0.596–0.836



Intervals use 1,000 bootstrap resamples clustered by player ID. All seasons for a hitter are resampled together, and all models are evaluated on the same draws. In 2026, each hitter contributes one record. These exploratory intervals are conditional on the observed sample; they do not incorporate full model-refitting uncertainty or resolve all temporal dependence.

Paired differences, including the direct M1–M3 comparison

Period	Population	Comparison	Difference	Exploratory 95% interval
2022–2025	PA-conditional	M2 − M1	+0.041	[+0.013, +0.067]
2022–2025	PA-conditional	M3 − M2	+0.009	[-0.006, +0.025]
2022–2025	PA-conditional	M3 − M1	+0.050	[+0.018, +0.078]
2022–2025	All followed	M2 − M1	+0.038	[+0.012, +0.065]
2022–2025	All followed	M3 − M2	+0.011	[-0.004, +0.026]
2022–2025	All followed	M3 − M1	+0.049	[+0.021, +0.078]
2026	PA-conditional	M2 − M1	+0.076	[+0.004, +0.149]
2026	PA-conditional	M3 − M2	+0.020	[-0.022, +0.067]
2026	PA-conditional	M3 − M1	+0.096	[+0.009, +0.181]
2026	All followed	M2 − M1	+0.063	[+0.001, +0.134]
2026	All followed	M3 − M2	+0.009	[-0.023, +0.048]
2026	All followed	M3 − M1	+0.072	[-0.002, +0.155]

The 2026 conditional M3 AUC is .782, with a wide individual interval of approximately .668–.893. Yet the paired M3 − M1 difference interval is approximately +.009–+.181. Overlap between individual AUC intervals alone cannot establish no model difference. The M3 − M2 interval includes zero. Much of the observed gain comes from adding **current skill levels**, rather than a large independent contribution from changes.

Retention sensitivity: All 36 combinations disclosed

Rise .010/.020/.030 × allowed loss .000/.010/.020 × next-season PA 100/200/300/400. The main definition is retained; the most favorable cell is not selected to construct the conclusion.

Period	View	Comparison	Definitions with positive difference	AUC difference: Minimum–maximum
2022–2025	All followed	M2 − M1	36/36	+0.017–+0.052
2022–2025	All followed	M3 − M2	33/36	-0.003–+0.017
2022–2025	All followed	M3 − M1	36/36	+0.020–+0.056
2022–2025	PA-conditional	M2 − M1	36/36	+0.008–+0.054
2022–2025	PA-conditional	M3 − M2	29/36	-0.009–+0.017
2022–2025	PA-conditional	M3 − M1	36/36	+0.008–+0.057
2026	All followed	M2 − M1	36/36	+0.011–+0.079
2026	All followed	M3 − M2	23/36	-0.012–+0.042
2026	All followed	M3 − M1	36/36	+0.014–+0.119
2026	PA-conditional	M2 − M1	36/36	+0.019–+0.102
2026	PA-conditional	M3 − M2	27/36	-0.012–+0.066
2026	PA-conditional	M3 − M1	36/36	+0.021–+0.168

M2-M1 is positive under 36/36 definitions in every period and population. M3-M2 changes direction under some definitions. Positive counts use full-precision AUC values: a difference can round to 0.000 yet be positive, so the development all-followed M3-M2 count remains 33/36. These overlapping definitions are not independent validations. All-followed success combines sufficient PA and retention; insufficient PA is decision failure, not proof of skill decline. This does not fully remove survival or opportunity selection from the wOBA model.

▼ Full AUC, denominators, and hitter counts for every definition

model	scope	view	rise	allowed_loss	target_pa_min	n	unique_players	success_n	M1_results	M2_levels	M3_changes
	development	all_followed_composite	0.01	0.00	100	381	289	96	0.714	0.742	0.759
	development	all_followed_composite	0.01	0.00	200	381	289	95	0.720	0.749	0.767
	development	all_followed_composite	0.01	0.00	300	381	289	81	0.703	0.736	0.746
	development	all_followed_composite	0.01	0.00	400	381	289	70	0.697	0.736	0.743
	development	all_followed_composite	0.01	0.01	100	381	289	128	0.684	0.716	0.730
	development	all_followed_composite	0.01	0.01	200	381	289	126	0.685	0.717	0.731
	development	all_followed_composite	0.01	0.01	300	381	289	110	0.654	0.690	0.696
	development	all_followed_composite	0.01	0.01	400	381	289	94	0.642	0.684	0.690
	development	all_followed_composite	0.01	0.02	100	381	289	166	0.693	0.716	0.725
	development	all_followed_composite	0.01	0.02	200	381	289	163	0.689	0.713	0.724
	development	all_followed_composite	0.01	0.02	300	381	289	146	0.659	0.684	0.689
	development	all_followed_composite	0.01	0.02	400	381	289	123	0.651	0.686	0.693
	development	all_followed_composite	0.02	0.00	100	292	243	64	0.725	0.757	0.771
	development	all_followed_composite	0.02	0.00	200	292	243	63	0.733	0.767	0.782
	development	all_followed_composite	0.02	0.00	300	292	243	54	0.726	0.761	0.767
	development	all_followed_composite	0.02	0.00	400	292	243	47	0.722	0.755	0.762
	development	all_followed_composite	0.02	0.01	100	292	243	87	0.695	0.731	0.741
	development	all_followed_composite	0.02	0.01	200	292	243	85	0.694	0.732	0.743
	development	all_followed_composite	0.02	0.01	300	292	243	75	0.676	0.715	0.718
	development	all_followed_composite	0.02	0.01	400	292	243	65	0.655	0.698	0.702
	development	all_followed_composite	0.02	0.02	100	292	243	116	0.701	0.718	0.725
	development	all_followed_composite	0.02	0.02	200	292	243	113	0.695	0.714	0.722
	development	all_followed_composite	0.02	0.02	300	292	243	102	0.678	0.696	0.698
	development	all_followed_composite	0.02	0.02	400	292	243	85	0.659	0.690	0.696
	development	all_followed_composite	0.03	0.00	100	221	193	34	0.709	0.757	0.760
	development	all_followed_composite	0.03	0.00	200	221	193	33	0.722	0.773	0.778
	development	all_followed_composite	0.03	0.00	300	221	193	29	0.732	0.779	0.776
	development	all_followed_composite	0.03	0.00	400	221	193	25	0.712	0.761	0.761
	development	all_followed_composite	0.03	0.01	100	221	193	51	0.660	0.707	0.710
	development	all_followed_composite	0.03	0.01	200	221	193	49	0.658	0.708	0.712
	development	all_followed_composite	0.03	0.01	300	221	193	44	0.649	0.697	0.695
	development	all_followed_composite	0.03	0.01	400	221	193	38	0.613	0.665	0.664
	development	all_followed_composite	0.03	0.02	100	221	193	75	0.692	0.713	0.716

development	all_followed_composite	0.03	0.02	200	221	193	72	0.683	0.707	0.712
development	all_followed_composite	0.03	0.02	300	221	193	67	0.674	0.693	0.694
development	all_followed_composite	0.03	0.02	400	221	193	55	0.641	0.674	0.680
development	conditional	0.01	0.00	100	344	260	96	0.735	0.764	0.781
development	conditional	0.01	0.00	200	328	245	95	0.742	0.774	0.790
development	conditional	0.01	0.00	300	277	213	81	0.752	0.777	0.789
development	conditional	0.01	0.00	400	226	176	70	0.761	0.785	0.793
development	conditional	0.01	0.01	100	344	260	128	0.710	0.743	0.757
development	conditional	0.01	0.01	200	328	245	126	0.711	0.746	0.759
development	conditional	0.01	0.01	300	277	213	110	0.712	0.739	0.748
development	conditional	0.01	0.01	400	226	176	94	0.718	0.742	0.749
development	conditional	0.01	0.02	100	344	260	166	0.728	0.753	0.762
development	conditional	0.01	0.02	200	328	245	163	0.728	0.754	0.763
development	conditional	0.01	0.02	300	277	213	146	0.746	0.760	0.766
development	conditional	0.01	0.02	400	226	176	123	0.768	0.784	0.792
development	conditional	0.02	0.00	100	264	219	64	0.746	0.780	0.794
development	conditional	0.02	0.00	200	250	206	63	0.759	0.795	0.809
development	conditional	0.02	0.00	300	209	176	54	0.775	0.805	0.812
development	conditional	0.02	0.00	400	169	146	47	0.792	0.818	0.821
development	conditional	0.02	0.01	100	264	219	87	0.719	0.758	0.768
development	conditional	0.02	0.01	200	250	206	85	0.724	0.765	0.774
development	conditional	0.02	0.01	300	209	176	75	0.735	0.769	0.771
development	conditional	0.02	0.01	400	169	146	65	0.740	0.773	0.772
development	conditional	0.02	0.02	100	264	219	116	0.735	0.754	0.760
development	conditional	0.02	0.02	200	250	206	113	0.737	0.757	0.764
development	conditional	0.02	0.02	300	209	176	102	0.765	0.773	0.774
development	conditional	0.02	0.02	400	169	146	85	0.781	0.800	0.802
development	conditional	0.03	0.00	100	203	177	34	0.718	0.768	0.770
development	conditional	0.03	0.00	200	191	166	33	0.734	0.788	0.790
development	conditional	0.03	0.00	300	161	139	29	0.766	0.811	0.806
development	conditional	0.03	0.00	400	130	114	25	0.765	0.812	0.806
development	conditional	0.03	0.01	100	203	177	51	0.670	0.719	0.721
development	conditional	0.03	0.01	200	191	166	49	0.672	0.724	0.725
development	conditional	0.03	0.01	300	161	139	44	0.691	0.735	0.729
development	conditional	0.03	0.01	400	130	114	38	0.675	0.721	0.713

development	conditional	0.03	0.02	100	203	177	75	0.709	0.731	0.733
development	conditional	0.03	0.02	200	191	166	72	0.706	0.730	0.732
development	conditional	0.03	0.02	300	161	139	67	0.741	0.753	0.749
development	conditional	0.03	0.02	400	130	114	55	0.737	0.763	0.760
final_2026	all_followed_composite	0.01	0.00	100	121	121	26	0.694	0.744	0.734
final_2026	all_followed_composite	0.01	0.00	200	121	121	24	0.661	0.715	0.702
final_2026	all_followed_composite	0.01	0.00	300	121	121	22	0.644	0.695	0.689
final_2026	all_followed_composite	0.01	0.00	400	121	121	19	0.615	0.674	0.671
final_2026	all_followed_composite	0.01	0.01	100	121	121	42	0.694	0.739	0.740
final_2026	all_followed_composite	0.01	0.01	200	121	121	38	0.660	0.710	0.706
final_2026	all_followed_composite	0.01	0.01	300	121	121	36	0.645	0.692	0.693
final_2026	all_followed_composite	0.01	0.01	400	121	121	32	0.621	0.664	0.665
final_2026	all_followed_composite	0.01	0.02	100	121	121	56	0.655	0.678	0.673
final_2026	all_followed_composite	0.01	0.02	200	121	121	51	0.629	0.655	0.643
final_2026	all_followed_composite	0.01	0.02	300	121	121	48	0.606	0.638	0.628
final_2026	all_followed_composite	0.01	0.02	400	121	121	41	0.551	0.581	0.578
final_2026	all_followed_composite	0.02	0.00	100	86	86	17	0.711	0.767	0.771
final_2026	all_followed_composite	0.02	0.00	200	86	86	16	0.684	0.743	0.746
final_2026	all_followed_composite	0.02	0.00	300	86	86	16	0.684	0.743	0.746
final_2026	all_followed_composite	0.02	0.00	400	86	86	14	0.637	0.707	0.706
final_2026	all_followed_composite	0.02	0.01	100	86	86	27	0.665	0.718	0.735
final_2026	all_followed_composite	0.02	0.01	200	86	86	25	0.648	0.711	0.720
final_2026	all_followed_composite	0.02	0.01	300	86	86	25	0.648	0.711	0.720
final_2026	all_followed_composite	0.02	0.01	400	86	86	22	0.603	0.664	0.668
final_2026	all_followed_composite	0.02	0.02	100	86	86	39	0.657	0.677	0.686
final_2026	all_followed_composite	0.02	0.02	200	86	86	36	0.649	0.678	0.676
final_2026	all_followed_composite	0.02	0.02	300	86	86	35	0.636	0.678	0.672
final_2026	all_followed_composite	0.02	0.02	400	86	86	30	0.553	0.592	0.587
final_2026	all_followed_composite	0.03	0.00	100	61	61	9	0.769	0.823	0.861
final_2026	all_followed_composite	0.03	0.00	200	61	61	8	0.726	0.785	0.828
final_2026	all_followed_composite	0.03	0.00	300	61	61	8	0.726	0.785	0.828
final_2026	all_followed_composite	0.03	0.00	400	61	61	7	0.685	0.765	0.804
final_2026	all_followed_composite	0.03	0.01	100	61	61	18	0.709	0.734	0.774
final_2026	all_followed_composite	0.03	0.01	200	61	61	16	0.679	0.718	0.753
final_2026	all_followed_composite	0.03	0.01	300	61	61	16	0.679	0.718	0.753

final_2026	all_followed_composite	0.03	0.01	400	61	61	14	0.634	0.672	0.701
final_2026	all_followed_composite	0.03	0.02	100	61	61	25	0.691	0.702	0.724
final_2026	all_followed_composite	0.03	0.02	200	61	61	22	0.671	0.695	0.706
final_2026	all_followed_composite	0.03	0.02	300	61	61	22	0.671	0.695	0.706
final_2026	all_followed_composite	0.03	0.02	400	61	61	19	0.599	0.620	0.633
final_2026	conditional	0.01	0.00	100	110	110	26	0.708	0.764	0.752
final_2026	conditional	0.01	0.00	200	92	92	24	0.697	0.764	0.755
final_2026	conditional	0.01	0.00	300	81	81	22	0.683	0.748	0.747
final_2026	conditional	0.01	0.00	400	69	69	19	0.678	0.759	0.759
final_2026	conditional	0.01	0.01	100	110	110	42	0.714	0.768	0.767
final_2026	conditional	0.01	0.01	200	92	92	38	0.718	0.787	0.791
final_2026	conditional	0.01	0.01	300	81	81	36	0.714	0.783	0.795
final_2026	conditional	0.01	0.01	400	69	69	32	0.736	0.802	0.814
final_2026	conditional	0.01	0.02	100	110	110	56	0.681	0.712	0.702
final_2026	conditional	0.01	0.02	200	92	92	51	0.705	0.743	0.736
final_2026	conditional	0.01	0.02	300	81	81	48	0.693	0.739	0.728
final_2026	conditional	0.01	0.02	400	69	69	41	0.669	0.710	0.709
final_2026	conditional	0.02	0.00	100	79	79	17	0.725	0.787	0.790
final_2026	conditional	0.02	0.00	200	66	66	16	0.712	0.777	0.790
final_2026	conditional	0.02	0.00	300	59	59	16	0.702	0.765	0.781
final_2026	conditional	0.02	0.00	400	50	50	14	0.710	0.790	0.806
final_2026	conditional	0.02	0.01	100	79	79	27	0.681	0.744	0.760
final_2026	conditional	0.02	0.01	200	66	66	25	0.687	0.763	0.782
final_2026	conditional	0.02	0.01	300	59	59	25	0.680	0.753	0.779
final_2026	conditional	0.02	0.01	400	50	50	22	0.708	0.781	0.807
final_2026	conditional	0.02	0.02	100	79	79	39	0.680	0.708	0.717
final_2026	conditional	0.02	0.02	200	66	66	36	0.716	0.748	0.756
final_2026	conditional	0.02	0.02	300	59	59	35	0.694	0.735	0.738
final_2026	conditional	0.02	0.02	400	50	50	30	0.682	0.717	0.723
final_2026	conditional	0.03	0.00	100	56	56	9	0.773	0.830	0.865
final_2026	conditional	0.03	0.00	200	46	46	8	0.747	0.812	0.862
final_2026	conditional	0.03	0.00	300	41	41	8	0.727	0.811	0.867
final_2026	conditional	0.03	0.00	400	35	35	7	0.735	0.837	0.903
final_2026	conditional	0.03	0.01	100	56	56	18	0.715	0.746	0.784
final_2026	conditional	0.03	0.01	200	46	46	16	0.712	0.762	0.804

final_2026	conditional	0.03	0.01	300	41	41	16	0.700	0.762	0.810
final_2026	conditional	0.03	0.01	400	35	35	14	0.718	0.772	0.827
final_2026	conditional	0.03	0.02	100	56	56	25	0.699	0.719	0.737
final_2026	conditional	0.03	0.02	200	46	46	22	0.727	0.761	0.769
final_2026	conditional	0.03	0.02	300	41	41	22	0.715	0.763	0.770
final_2026	conditional	0.03	0.02	400	35	35	19	0.707	0.743	0.760

Hanley & McNeil (1982), Radiology 143:29–36, DOI 10.1148/radiology.143.1.7063747 discusses sampling variability in AUC. Paired AUC comparisons on the same cases are treated separately in Hanley & McNeil (1983), DOI 10.1148/radiology.148.3.6878708. This study uses player-cluster bootstrap, not those approximation formulas.

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05 · REVISED DECISION LISTS

Separate baseline forecasts from unvalidated signal hypotheses

The original discovery proxy required wOBA ≤ the league PA-weighted mean+.020, HR ≤25, and no season in the preceding 2 years with 200PA and wOBA ≥.360. It does not measure attention or acquisition price. Adding **observed year-to-year wOBA growth <.020** separates apparent breakouts from less-visible change hypotheses. This revision followed inspection of results and remains exploratory.

M1 baseline: 5 statistical comparators

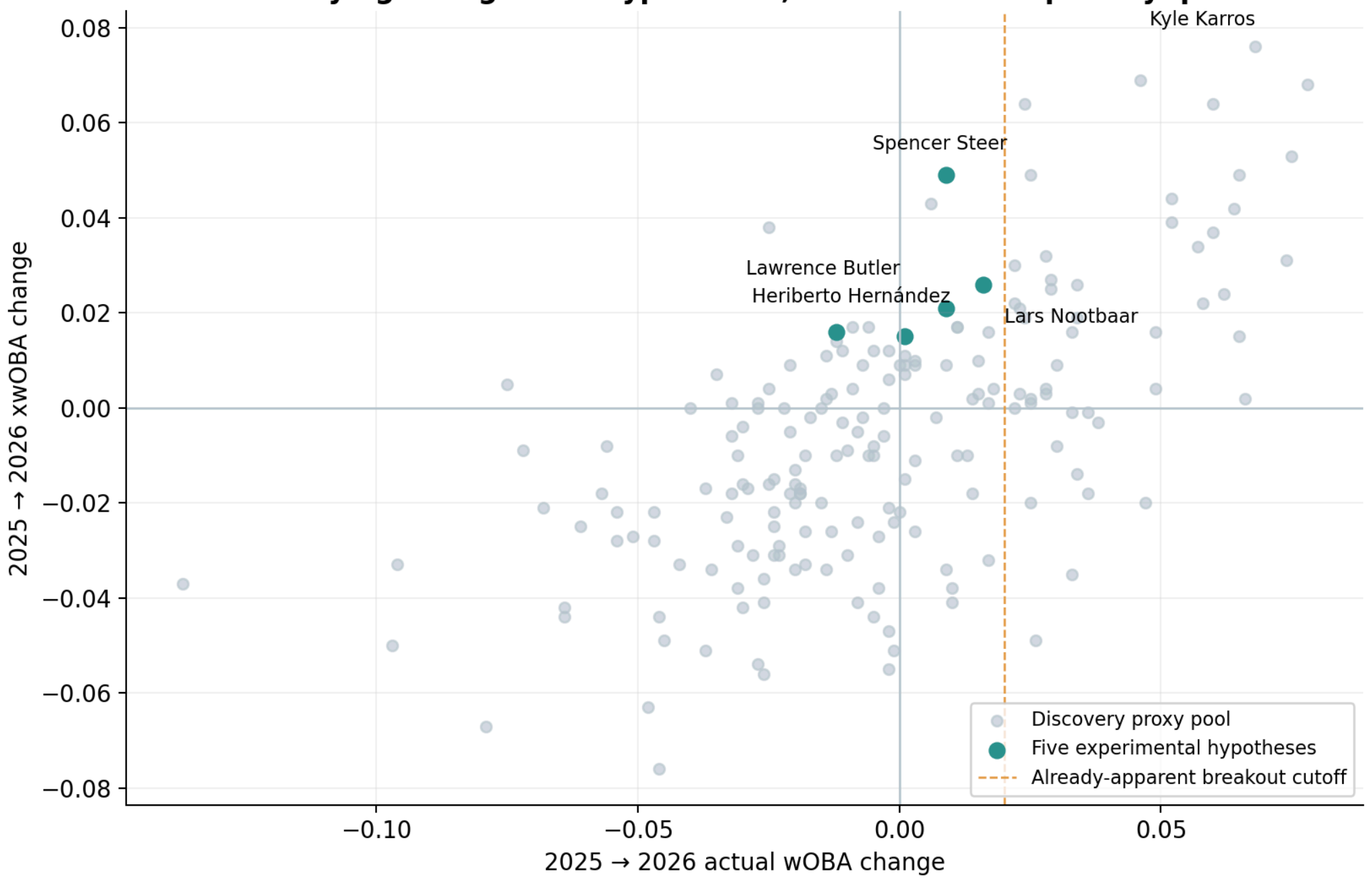
Hitter	2026 wOBA	2026 xwOBA	2027 M1 forecast
Wyatt Langford	0.333	0.319	0.338
Vinnie Pasquantino	0.331	0.338	0.335
Wilyer Abreu	0.335	0.346	0.333
Cody Bellinger	0.336	0.335	0.332
Trent Grisham	0.311	0.337	0.332

Under the same historical conditions, M1-only produced more observed growth successes. This small exploratory sample does not establish general policy superiority and is not independent 2027 evidence. Familiar names and substantial contracts may appear; this is not a low-budget acquisition recommendation. Contract terms and acquisition feasibility require separate review.

Separate experiment: 5 less-visible change hypotheses

Display order	Hitter	woba	xwoba	wOBA change	xwOBA change	2027 M1 forecast	Hypothesis to investigate
1	Heriberto Hernández	0.329	0.347	-0.012	+0.016	0.329	Better contact quality despite declining results
2	Lawrence Butler	0.319	0.326	+0.009	+0.021	0.324	Contact and swing-selection gains rather than power gains
3	Spencer Steer	0.325	0.342	+0.009	+0.049	0.322	Contact-quality gains beyond modest result improvement
4	Lars Nootbaar	0.320	0.343	+0.016	+0.026	0.321	Strong contact level and swing selection
5	Tyler Stephenson	0.320	0.326	+0.001	+0.015	0.312	Contact and swing-selection recovery

The existing signal rule is retained: current xwOBA ≥.320, change ≥.010, and at least 2 of EV +.5mph / barrel +1pp / chase −1pp / whiff −1pp. Display order follows M1 forecasts, but **this ordering has not established superiority over simple or random selection.** Neither the original rule nor the revised 5-hitter rule established superior growth yield. Signal hypotheses do not replace M1 baseline review.



▼ Separate retention-monitoring group: 8 apparent breakouts

Hitter	2025 PA	wOBA change	xwOBA change	2026 wOBA
JJ Bleday	344.0	+0.024	+0.064	0.326
Coby Mayo	294.0	+0.025	+0.049	0.328
Kyle Karros	156.0	+0.068	+0.076	0.336
Casey Schmitt	348.0	+0.029	+0.027	0.336
Nathaniel Lowe	609.0	+0.028	+0.032	0.329
Drew Gilbert	109.0	+0.060	+0.064	0.320
Brady House	274.0	+0.065	+0.049	0.313
Luke Raley	219.0	+0.022	+0.030	0.313

Kyle Karros already showed a wOBA increase of +.068, with a small preceding sample of 156PA. He was moved from the main growth-hypothesis list to retention monitoring. This group is not described as hidden, undervalued hitters.

06 · EVIDENCE & OPPOSITION

Evidence and verification tasks for five hypotheses

Change metrics do not guarantee growth. The error ranges below use the 10–90 percentiles of historical annual residuals; they are not guaranteed future 80% intervals. Forecasts are conditional on sufficient next-season MLB playing time.

M1 uses 9 traditional variables; M2 adds 6 skill levels; M3 adds 6 changes (9/15/21 inputs). Hard-hit, K%, BB%, and ISO already enter M1/M2. The cards restate them as evidence; only mean launch angle is newly displayed, and it is excluded from the models.

EXPERIMENTAL HYPOTHESIS 1 · Selection superiority unproven

Heriberto Hernández

Listed team: Miami Marlins · July age 26.5 · 2026 505PA / prior 294PA

wOBA **0.341 → 0.329** xwOBA **0.331 → 0.347**

Hypothesis: Better contact quality despite declining results

Supporting evidence: wOBA fell from .341→.329 while xwOBA rose from .331→.347. EV and barrel improved alongside lower whiff.

Counterevidence: Chase worsened by +2.4pp, and whiff remained high at 32.5%. M1 forecasts .329, without a large improvement.

EV +1.3mph · barrel +1.7pp · chase +2.4pp · whiff -2.7pp

2027 M1 forecast **0.329**, change +0.000 · Historical error range 0.288–0.365

Video-review task: Review strong contact by pitch type and location, and whether poorer chase decisions are concentrated in specific situations.

Unverified: 2027 salary, options, acquisition compensation, role, and health. These are not success probabilities or value-discount estimates.

Additional observations · 2025 → 2026
Hard-hit 45.4% → 48.3% · K 26.2% → 26.5% · BB 10.5% → 10.3%
ISO 0.172 → 0.214 · Mean launch angle 14.2° → 13.2°

Interpretation: Hard-hit and ISO increased, but K% and BB% did not improve. Lower whiff did not translate into a lower overall strikeout rate. Investigate stronger contact separately from contact by pitch type.

EXPERIMENTAL HYPOTHESIS 2 · Selection superiority unproven

Lawrence Butler

Listed team: Athletics · July age 26.0 · 2026 540PA / prior 630PA

wOBA **0.310 → 0.319** xwOBA **0.305 → 0.326**

Hypothesis: Contact and swing-selection gains rather than power gains

Supporting evidence: Chase −1.5pp, whiff −6.2pp, and xwOBA +.021; observed wOBA rose only +.009.

Counterevidence: EV −0.8mph and barrel −0.6pp. Whether better contact offsets reduced power is uncertain.

EV -0.8mph · barrel -0.6pp · chase -1.5pp · whiff -6.2pp

2027 M1 forecast **0.324**, change +0.005 · Historical error range 0.282–0.359

Video-review task: Determine whether more contact means more weak contact or better swing selection on favorable locations.

Unverified: 2027 salary, options, acquisition compensation, role, and health. These are not success probabilities or value-discount estimates.

Additional observations · 2025 → 2026
Hard-hit 44.4% → 44.8% · K 28.4% → 25.7% · BB 9.4% → 11.3%
ISO 0.170 → 0.156 · Mean launch angle 9.2° → 7.7°

Interpretation: Lower K% and higher BB% support the contact and swing-selection hypothesis. ISO and mean launch angle declined. Distinguish contact gains from power gains, and review in-zone contact and weak-contact share.

EXPERIMENTAL HYPOTHESIS 3 · Selection superiority unproven

Spencer Steer

Listed team: Cincinnati Reds · July age 28.6 · 2026 384PA / prior 568PA

wOBA **0.316 → 0.325** xwOBA **0.293 → 0.342**

Hypothesis: Contact-quality gains beyond modest result improvement

Supporting evidence: wOBA .316→.325 versus xwOBA .293→.342, EV +2.4mph, and barrel +4.8pp.

Counterevidence: Chase +1.4pp and whiff +0.3pp indicate no swing-selection or contact improvement. M1 forecasts .322.

EV +2.4mph · barrel +4.8pp · chase +1.4pp · whiff +0.3pp

2027 M1 forecast **0.322**, change −0.003 · Historical error range 0.280–0.357

Video-review task: Check whether contact-quality gains are limited to specific pitch types or pulled balls. Their repeatability is the investigation question.

Unverified: 2027 salary, options, acquisition compensation, role, and health. These are not success probabilities or value-discount estimates.

Additional observations · 2025 → 2026
Hard-hit 33.2% → 39.1% · K 22.7% → 22.4% · BB 9.0% → 9.1%
ISO 0.173 → 0.183 · Mean launch angle 18.8° → 16.9°

Interpretation: Higher hard-hit and ISO support improved contact quality. A lower mean launch angle does not establish a better launch-angle distribution. Review sweet-spot% and repeatability of pulled airborne contact.

EXPERIMENTAL HYPOTHESIS 4 · Selection superiority unproven

Lars Nootbaar

Listed team: Arizona Diamondbacks · July age 28.8 · 2026 334PA / prior 583PA

wOBA **0.304 → 0.320** xwOBA **0.317 → 0.343**

Hypothesis: Strong contact level and swing selection

Supporting evidence: xwOBA .343 and EV 92.0mph. EV +0.7mph, chase −1.3pp, and whiff −1.8pp show relatively consistent directions.

Counterevidence: 334PA after injury recovery and a trade; barrel -0.1pp. M1 .321 vs M2 .334 shows input sensitivity.

EV +0.7mph · barrel -0.1pp · chase -1.3pp · whiff -1.8pp

2027 M1 forecast **0.321**, change +0.001 · Historical error range 0.279–0.356

Season context: Returned June 5 after surgery on both heels; traded to Arizona August 3. Savant: STL 48G/183PA/.234/3HR + Arizona 42G/151PA/5HR = 334PA; season wOBA .320. Review health, opportunity, and within-season signal consistency. [P1-P3]

Unverified: 2027 salary, options, acquisition compensation, role, and health. These are not success probabilities or value-discount estimates.

Additional observations · 2025 → 2026
Hard-hit 50.0% → 49.5% · K 20.4% → 21.0% · BB 11.0% → 13.8%
ISO 0.127 → 0.140 · Mean launch angle 16.5° → 9.2°

Interpretation: BB% increased, but K% also rose slightly and hard-hit declined. Check whether the lower mean launch angle reflects more ground balls. The 334PA sample, role, and health add uncertainty.

EXPERIMENTAL HYPOTHESIS 5 · Selection superiority unproven

Tyler Stephenson

Listed team: Cincinnati Reds · July age 29.9 · 2026 468PA / prior 342PA

wOBA **0.319 → 0.320** xwOBA **0.311 → 0.326**

Hypothesis: Contact and swing-selection recovery

Supporting evidence: Chase −2.9pp and whiff −4.1pp. xwOBA +.015 exceeded observed wOBA growth of +.001.

Counterevidence: Barrel −2.1pp; M1 forecasts .312. Batting metrics alone do not measure a catcher's total value.

EV +0.3mph · barrel -2.1pp · chase -2.9pp · whiff -4.1pp

2027 M1 forecast **0.312**, change −0.008 · Historical error range 0.271–0.348

Video-review task: Review improved swing selection alongside reduced power. Catcher defense, health, and playing-time burden require separate evaluation.

Unverified: 2027 salary, options, acquisition compensation, role, and health. These are not success probabilities or value-discount estimates.

Additional observations · 2025 → 2026
Hard-hit 49.2% → 44.1% · K 33.9% → 27.4% · BB 10.8% → 10.5%
ISO 0.190 → 0.163 · Mean launch angle 12.9° → 18.0°

Interpretation: Lower K% supports contact recovery, but BB%, hard-hit, and ISO declined. Determine whether the higher mean launch angle reflects productive airborne contact or more fly balls and popups.

Additional observations use the existing 2025 and 2026 Custom files and processed CSVs. Mean launch angle is a summary; it cannot replace sweet-spot% or batted-ball type shares. In-zone contact, pitch-type and handedness denominators, bat tracking, park context, health, and role were not measured in this package. They are investigation questions, not established causes of improvement.

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07 · TEST THE ACTUAL SELECTION POLICY

Who is selected under the same budget?

For each target year in 2022–2026, screens and rankings used only that year's inputs. The original policy allowed at most 13 hitters and the revised policy at most 5; k fell when fewer hitters passed the screen. **Every comparator within a year and policy uses the same k and base candidate pool**. The original policy totals 53 slots and the revised policy 22 slots. Reselecting a hitter in another season creates another slot; these are not 53 or 22 independent hitters.

Growth success = next-season PA ≥200 and wOBA increase ≥+.020. Usable bat = PA ≥200 and wOBA ≥.330. All selected hitters remain in denominators, including those with insufficient PA or no subsequent record. Comparing screen→M1 with screen→xwOBA change tests **ordering within the screened set**; comparison with M1-only and other methods evaluates the **complete selection policy, including the screen**.

Policy	Method	Investigation slots	Growth successes	Usable-bat successes	Reached 200PA
Original: Max. 13	Signal screen→M1	53	7	12	47
Original: Max. 13	Same screen→xwOBA change	53	10	9	44
Original: Max. 13	M1-only	53	13	19	47
Original: Max. 13	Current wOBA	53	7	12	43
Original: Max. 13	Current xwOBA	53	12	17	51
Original: Max. 13	xwOBA change	53	13	10	42
Original: Max. 13	Random expectation	53	11.2	7.2	36.0

Revised: Max. 5	Signal screen→M1	22	2	6	19
Revised: Max. 5	Same screen→xwOBA change	22	4	5	20
Revised: Max. 5	M1-only	22	7	9	21
Revised: Max. 5	Current wOBA	22	4	6	19
Revised: Max. 5	Current xwOBA	22	4	8	22
Revised: Max. 5	xwOBA change	22	6	5	19
Revised: Max. 5	Random expectation	22	5.2	3.2	14.7

Why AUC and selection yield differ: Targets, populations, and loss points

Policy	Method	All slots	Unique hitters	Below 200PA	200PA reached; no growth	Growth successes	Growth among available
Original: Max. 13	Signal screen → M1	53	51	6	40	7	7/47 (14.9%)
Original: Max. 13	M1-only	53	46	6	34	13	13/47 (27.7%)
Revised: Max. 5	Signal screen → M1	22	22	3	17	2	2/19 (10.5%)
Revised: Max. 5	M1-only	22	22	1	14	7	7/21 (33.3%)

Slots = below 200PA + sufficient PA without growth + growth success. Under the original rule, both methods reached 200PA in 47 slots, yet growth successes were 7 versus 13. Playing-time shortfalls alone cannot explain that gap. Under the revised rule, below-200PA counts were 3 versus 1 and available-but-no-growth counts were 17 versus 14. This locates losses; it does not identify injury, role, or skill as their cause.

AUC ranks **retention** of gains among hitters whose wOBA already rose ≥+.020, allowing next-season loss ≤.010. Policy success instead requires selecting future **growth** of at least +.020 from a restricted discovery pool. The revised rule excludes already-apparent breakouts. Overall ranking discrimination does not guarantee top-k yield, calibrated probabilities, or screen effectiveness. PA conditioning differs, but so do the target, candidate population, and selection task.

The original two policies share 5 player-season selections; the revised policies share 1. Repeated hitters and shared selections prevent using an independent two-proportion z-test as confirmatory evidence. Only 5 seasons are observed, and some rules were revised after results. Player-season records and overlap tables are disclosed; the frozen 2027 evaluation will assess prospective performance. Files: supplement/policy_loss_decomposition.csv, policy_selection_overlap.csv.

Growth success: Compare the random distribution, not only its mean

Policy	Method	Slots	Growth successes	Random mean	Standard deviation	2.5% quantile	97.5% quantile	Two-sided tail
Original: Max. 13	Signal screen → M1	53	7	11.20	2.86	6	17	0.164
Original: Max. 13	M1-only	53	13	11.20	2.86	6	17	0.490
Revised: Max. 5	Signal screen → M1	22	2	5.18	1.95	2	9	0.126
Revised: Max. 5	M1-only	22	7	5.18	1.95	2	9	0.440

The screen fell below the random mean and M1 exceeded it, but all four success counts lie within their central 95% random-selection ranges. Neither method was clearly distinguishable from this benchmark; failure to detect a difference does not establish equivalence. Usable-bat yield of 19 versus a random expectation of 7.2 is a separate outcome. M1 ranks absolute projected wOBA, which aligns with the ≥.330 target. That result does not establish additional growth or signal-screen superiority. Alignment between selection and outcome is neither a guarantee of future success nor an evaluation error.

For each year, a hypergeometric distribution describes choosing k hitters without replacement from the same candidate pool. These distributions are convolved under independent random selections across years; observed player outcomes are not assumed independent. The two-sided tail sums outcomes with probability mass no greater

than the observed count's mass. Historical rule revisions, multiple comparisons, and player dependence prevent treating this benchmark as a general policy-effect test or confidence interval. CSVs: supplement/random_growth_benchmark.csv, random_growth_distribution.csv.

The signal rule produced fewer observed growth successes than M1-only or random expectation. Usable-bat yield exceeded random expectation but fell below M1-only. These differences do not prove general policy inferiority. Growth and batting level are distinct targets. Evidence is insufficient to change the default M1 policy; a more favorable ordering is not selected after results.

The random baseline uses the exact expectation for choosing k hitters without replacement from the same base pool. Annual hypergeometric 95% selection ranges appear in the CSV and detailed table below. They describe random selection, not confidence intervals for a statistical effect. With only 5 seasons, repeated hitters, and retrospective definition changes, these results are not confirmatory evidence of policy superiority.

▼ Full denominators and success counts by policy and year

year	policy	method	base_pool_n	screen_pool_n	selected_n	growth_successes	usable_successes	available_n	random_growth_low	random_growth_high
2022	original_13	Signal screen→M1	122	8	8	1.00	1.00	7.00	NaN	NaN
2022	original_13	Same screen→xwOBA change	122	8	8	1.00	1.00	7.00	NaN	NaN
2022	original_13	M1-only	122	8	8	3.00	2.00	5.00	NaN	NaN
2022	original_13	Current wOBA	122	8	8	2.00	2.00	8.00	NaN	NaN
2022	original_13	Current xwOBA	122	8	8	2.00	2.00	7.00	NaN	NaN
2022	original_13	xwOBA change	122	8	8	0.00	0.00	5.00	NaN	NaN
2022	original_13	Random expectation	122	8	8	1.51	0.92	5.44	0.0	4.0
2022	revised_hidden_5	Signal screen→M1	105	2	2	0.00	0.00	2.00	NaN	NaN
2022	revised_hidden_5	Same screen→xwOBA change	105	2	2	0.00	0.00	2.00	NaN	NaN
2022	revised_hidden_5	M1-only	105	2	2	1.00	1.00	2.00	NaN	NaN
2022	revised_hidden_5	Current wOBA	105	2	2	0.00	0.00	2.00	NaN	NaN
2022	revised_hidden_5	Current xwOBA	105	2	2	1.00	1.00	2.00	NaN	NaN
2022	revised_hidden_5	xwOBA change	105	2	2	0.00	0.00	2.00	NaN	NaN
2022	revised_hidden_5	Random expectation	105	2	2	0.38	0.21	1.33	0.0	2.0
2023	original_13	Signal screen→M1	173	6	6	1.00	2.00	5.00	NaN	NaN
2023	original_13	Same screen→xwOBA change	173	6	6	1.00	2.00	5.00	NaN	NaN
2023	original_13	M1-only	173	6	6	2.00	3.00	5.00	NaN	NaN
2023	original_13	Current wOBA	173	6	6	2.00	3.00	4.00	NaN	NaN
2023	original_13	Current xwOBA	173	6	6	2.00	3.00	6.00	NaN	NaN
2023	original_13	xwOBA change	173	6	6	1.00	3.00	4.00	NaN	NaN
2023	original_13	Random expectation	173	6	6	1.56	0.97	3.95	0.0	4.0
2023	revised_hidden_5	Signal screen→M1	150	6	5	0.00	1.00	4.00	NaN	NaN
2023	revised_hidden_5	Same screen→xwOBA change	150	6	5	1.00	2.00	5.00	NaN	NaN
2023	revised_hidden_5	M1-only	150	6	5	2.00	2.00	4.00	NaN	NaN
2023	revised_hidden_5	Current wOBA	150	6	5	2.00	2.00	4.00	NaN	NaN

2023	revised_hidden_5	Current xwOBA	150	6	5	1.00	2.00	5.00	NaN	NaN
2023	revised_hidden_5	xwOBA change	150	6	5	1.00	2.00	4.00	NaN	NaN
2023	revised_hidden_5	Random expectation	150	6	5	1.33	0.73	3.23	0.0	3.0
2024	original_13	Signal screen→M1	170	16	13	1.00	2.00	10.00	NaN	NaN
2024	original_13	Same screen→xwOBA change	170	16	13	1.00	1.00	9.00	NaN	NaN
2024	original_13	M1-only	170	16	13	2.00	3.00	13.00	NaN	NaN
2024	original_13	Current wOBA	170	16	13	1.00	2.00	11.00	NaN	NaN
2024	original_13	Current xwOBA	170	16	13	1.00	4.00	12.00	NaN	NaN
2024	original_13	xwOBA change	170	16	13	1.00	0.00	10.00	NaN	NaN
2024	original_13	Random expectation	170	16	13	1.99	1.07	9.33	0.0	5.0
2024	revised_hidden_5	Signal screen→M1	124	7	5	0.00	1.00	3.00	NaN	NaN
2024	revised_hidden_5	Same screen→xwOBA change	124	7	5	0.00	1.00	3.00	NaN	NaN
2024	revised_hidden_5	M1-only	124	7	5	1.00	1.00	5.00	NaN	NaN
2024	revised_hidden_5	Current wOBA	124	7	5	1.00	2.00	4.00	NaN	NaN
2024	revised_hidden_5	Current xwOBA	124	7	5	1.00	3.00	5.00	NaN	NaN
2024	revised_hidden_5	xwOBA change	124	7	5	2.00	1.00	4.00	NaN	NaN
2024	revised_hidden_5	Random expectation	124	7	5	0.93	0.52	3.47	0.0	3.0
2025	original_13	Signal screen→M1	201	17	13	1.00	4.00	13.00	NaN	NaN
2025	original_13	Same screen→xwOBA change	201	17	13	2.00	3.00	12.00	NaN	NaN
2025	original_13	M1-only	201	17	13	2.00	6.00	12.00	NaN	NaN
2025	original_13	Current wOBA	201	17	13	0.00	2.00	9.00	NaN	NaN
2025	original_13	Current xwOBA	201	17	13	2.00	4.00	13.00	NaN	NaN
2025	original_13	xwOBA change	201	17	13	4.00	4.00	11.00	NaN	NaN
2025	original_13	Random expectation	201	17	13	2.59	2.13	8.21	0.0	5.0
2025	revised_hidden_5	Signal screen→M1	170	8	5	0.00	2.00	5.00	NaN	NaN
2025	revised_hidden_5	Same screen→xwOBA change	170	8	5	0.00	2.00	5.00	NaN	NaN
2025	revised_hidden_5	M1-only	170	8	5	1.00	2.00	5.00	NaN	NaN

2025	revised_hidden_5	Current wOBA	170	8	5	0.00	1.00	4.00	NaN	NaN
2025	revised_hidden_5	Current xwOBA	170	8	5	0.00	1.00	5.00	NaN	NaN
2025	revised_hidden_5	xwOBA change	170	8	5	0.00	2.00	4.00	NaN	NaN
2025	revised_hidden_5	Random expectation	170	8	5	1.00	0.82	3.09	0.0	3.0
2026	original_13	Signal screen→M1	183	22	13	3.00	3.00	12.00	NaN	NaN
2026	original_13	Same screen→xwOBA change	183	22	13	5.00	2.00	11.00	NaN	NaN
2026	original_13	M1-only	183	22	13	4.00	5.00	12.00	NaN	NaN
2026	original_13	Current wOBA	183	22	13	2.00	3.00	11.00	NaN	NaN
2026	original_13	Current xwOBA	183	22	13	5.00	4.00	13.00	NaN	NaN
2026	original_13	xwOBA change	183	22	13	7.00	3.00	12.00	NaN	NaN
2026	original_13	Random expectation	183	22	13	3.55	2.13	9.09	1.0	7.0
2026	revised_hidden_5	Signal screen→M1	146	14	5	2.00	2.00	5.00	NaN	NaN
2026	revised_hidden_5	Same screen→xwOBA change	146	14	5	3.00	0.00	5.00	NaN	NaN
2026	revised_hidden_5	M1-only	146	14	5	2.00	3.00	5.00	NaN	NaN
2026	revised_hidden_5	Current wOBA	146	14	5	1.00	1.00	5.00	NaN	NaN
2026	revised_hidden_5	Current xwOBA	146	14	5	1.00	1.00	5.00	NaN	NaN
2026	revised_hidden_5	xwOBA change	146	14	5	3.00	0.00	5.00	NaN	NaN
2026	revised_hidden_5	Random expectation	146	14	5	1.54	0.89	3.56	0.0	4.0

How many selections change when average prediction errors barely differ?

comparison	Target year	M1→M2 replacements	M1→M3 replacements	M2→M3 replacements
	2022	7	6	1
	2023	4	4	2
	2024	4	5	3
	2025	6	5	4
	2026	6	5	3
	2027_forecast	5	5	2

Top15 lists were compared within the same discovery pool. In 2026, M2→M3 worsened RMSE by approximately .00008 while replacing 3 hitters. In the frozen 2027 forecasts, M2→M3 replaced 2 hitters; M1→M2 and M1→M3 each replaced 5. Small average error differences can change selections near a ranking cutoff, but **replacement itself is not better performance or economic benefit.** Top5 results, average rank changes, and added/excluded names are also disclosed in CSVs.

Freeze the 2027 evaluation rules before outcomes

After the complete 2027 regular season, evaluate actual PA and wOBA for **all 298 frozen hitters**. Report primary RMSE and MAE, secondary retention under .020/.010/200, all 36 sensitivity definitions, all-followed outcomes including insufficient PA, paired AUC differences, and equal-budget policy comparisons. Models, candidates, and criteria will not change to fit 2027 outcomes. Any change receives a separate exploratory version.

Original forecast and evaluation-plan timestamps are distinct. Observed 2026 results are not retrospectively relabeled independent validation. All actual 2027 outcomes remain pending.

▼ Full frozen 2027 evaluation protocol

```
# Signals for Next Season — prospective 2027 evaluation protocol

Locked on 2026-10-02, after observing 2022–2026 exploratory results, before any 2027 regular-season outcomes. This is a dated project commitment, not an externally registered preregistration. A SHA256 manifest binds this document, the frozen predictions and the evaluation code. Never overwrite the first freeze silently.

## Questions and status

Primary: How well do frozen models forecast next-season batting quality?

Secondary, preserved as secondary: Do current skill levels and year-to-year changes discriminate retention of an apparent breakout?

Decision evaluation: Does a fixed skill-change investigation screen yield more future growth or usable bats than equal-budget simple or random selection? The decision screen was designed after historical results. Its 2022–2026 evaluation is exploratory; 2027 is the first prospective evaluation of this revised policy.

No within-season early-detection, causal development, media obscurity, acquisition availability or market-undervaluation claim is made.

## Cohort and frozen forecasts

Use full 2026 regular-season inputs only. Main eligible cohort: predictor PA >=200 and immediately preceding PA >=100. Preserve all 298 eligible hitters and all M1/M2/M3 forecasts, not just selected names. M1 remains the adopted projection policy selected using already-observed historical results. Ridge alpha=10 and all feature definitions remain unchanged. Fit using completed target seasons through 2026, with target PA >=200. Freeze before outcomes; do not choose a new model using 2027 results and then present it as the original policy.

## Evaluation after the complete 2027 regular season

Require full-season 2027 MLB actual PA and wOBA, excluding postseason, plus source and retrieval metadata. Explicitly confirm that the season is complete. A player absent from complete MLB totals receives PA=0 and unavailable wOBA. Missingness in an incomplete feed is not a zero outcome.

Primary continuous metrics: equal-player RMSE and MAE among target PA >=200, with n and unique players. Compare B0/current wOBA, M1, M2, M3 using identical target observations. PA-weighted RMSE is supplementary. Predicting playing time is outside this model.

Secondary apparent-breakout cohort: 2026 wOBA increased by >=.020 over 2025. Conditional sustained label: 2027 actual wOBA >= 2026 actual wOBA minus .010, among 2027 PA >=200. Discrimination score: frozen model projection minus (2026 wOBA minus .010). Report ROC AUC, accuracy, balanced accuracy, counts and paired AUC differences M2–M1, M3–M2, M3–M1. Player-cluster bootstrap 1,000 resamples, seed 4427; resample all model scores together. Report each AUC interval as well. Also report the all-followed composite: PA >=200 AND retained wOBA, treating
```

insufficient PA as an unsuccessful decision outcome, not evidence of skill deterioration.

Sensitivity, without selecting the best definition: rise thresholds .010/.020/.030; allowed subsequent losses .000/.010/.020; target PA cutoffs 100/200/300/400. Publish all 36 combinations, both conditional and all-followed composite, all models. Label the original .020/.010/200 definition as main; do not replace it using future results.

Investigation policy and comparators

Base discovery proxy: current wOBA <= PA-weighted all-hitter mean +.020; HR <=25; no preceding-two-year season with PA >=200 and wOBA >=.360. These are production proxies, not price/attention measurements.

Revised less-visible-growth base pool also requires current minus prior wOBA <.020. Already-apparent breakouts are a separate monitoring group, not hidden-growth recommendations.

Signal screen: current xwOBA >=.320; year-to-year xwOBA change >=.010; at least two of EV +.5mph, barrel +1 percentage point, chase −1 point, whiff −1 point. These cutoffs are an analyst screen, not optimized or learned breakout probabilities.

Investigation budget is at most five hitters. Select min(5, screened count) by frozen M1 projected wOBA descending, MLB player ID ascending as tie-break. Freeze the exact 2027 IDs. Compare same k against current-wOBA, current-xwOBA, delta-xwOBA and M1-only rankings within the same less-visible-growth base pool. A within-screen delta-xwOBA ranking is an ordering comparator. A random baseline draws k without replacement from the same base pool; expected yield and hypergeometric selection ranges describe random choice, not inferential confidence. Do not change denominators or pool after observing outcomes.

Decision outcomes, all selected hitters in denominators:
- Growth: 2027 PA >=200 AND wOBA improves by >=.020 over 2026.
- Usable bat: 2027 PA >=200 AND wOBA >=.330.
- Report PA availability separately, and average wOBA only with its available count.

Also retain the original 13-name screen as an explicitly exploratory legacy comparison, with budget capped at 13 and its own original broad base pool. Do not mix the two decision denominators.

Report all results

Report successes and failures, all prespecified comparisons and sensitivity cells, even when weaker than random/simple selection. Bootstrap intervals are exploratory; limited years and remaining dependence prevent strong universal claims. Near-threshold values use recorded precision (wOBA rounded to .001). No invented future outcomes, unvalidated calibrated breakout probabilities or club ROI.

If a rule changes after this freeze, publish a new dated exploratory analysis and preserve this frozen version. Original forecast timestamp and revised policy-lock timestamp are distinct. Do not retrospectively rename observed 2026 data an independent holdout.

Protocol SHA256: 71d75971d1b7ecbae840f35258173fdd73812768e96c5f10c3c0711a05f917c3
Hashes of the 298 forecasts, policy IDs, and evaluation code are also frozen in 2027_evaluation_lock.json. This is not a claim of externally registered preregistration.

Decision use and the Moneyball scope

1. Start with traditional forecasts such as M1 as the baseline.
2. Use change metrics to investigate conflicting player-level signals rather than force promotion or exclusion.
3. Record the unvalidated signal list as a separate experiment and compare 2027 outcomes with M1, simple rankings, and random selection.
4. Review salary, options, acquisition feasibility, compensation costs, playing-time role, health, and defense before making economic decisions.
   Without consistent 2027 pricing data, this report does not estimate market undervaluation or ROI.

Observed growth is not identified skill development

The +.020 growth outcome can include rebound from unusually low results through regression to the mean. This study does not separate luck, environment, and skill, and does not establish regression to the mean as the explanation for policy differences. Success means observed batting improvement, not a causal effect of behavior change. Future diagnostics can show the current xwOBA−wOBA gap, multi-year history, and M1 projected rebound without changing the original growth label. Any adjusted skill outcome must be defined as a separate exploratory analysis.

APPENDIX A · CONSERVATIVE FORECAST RANKING

M1 forecast Top15: Separate from signal investigations

Rank	Hitter	2026 wOBA	2027 forecast	Projected change	Bootstrap Top15 share	Use
1	Wyatt Langford	0.333	0.338	+0.005	100%	Forecast comparator
2	Vinnie Pasquantino	0.331	0.335	+0.004	100%	Forecast comparator

3	Wilyer Abreu	0.335	0.333	-0.002	100%	Forecast comparator
4	Cody Bellinger	0.336	0.332	-0.004	100%	Contract review required · Established name
5	Trent Grisham	0.311	0.332	+0.021	100%	Forecast comparator
6	Nolan Schanuel	0.328	0.331	+0.003	100%	Forecast comparator
7	Alejandro Kirk	0.333	0.330	-0.003	100%	Contract review required · Established name
8	William Contreras	0.330	0.330	+0.000	100%	Forecast comparator
9	Maikel Garcia	0.318	0.330	+0.012	100%	Forecast comparator
10	Heriberto Hernández	0.329	0.329	+0.000	98%	Forecast comparator
11	Julio Rodríguez	0.310	0.327	+0.017	91%	Contract review required · Established name
12	Steven Kwan	0.334	0.327	-0.007	74%	Forecast comparator
13	Matt Shaw	0.325	0.326	+0.001	77%	Forecast comparator
14	Spencer Horwitz	0.319	0.325	+0.006	33%	Forecast comparator
15	Gleyber Torres	0.328	0.325	-0.003	43%	Forecast comparator

Bootstrap Top15 share is the proportion of 150 refits, with training hitters resampled as clusters, in which a hitter entered the Top15. A value of 100% is not a 100% breakout probability. Future outcome uncertainty is much larger; small rank differences should not be overinterpreted.

Example: Langford ranks 1st under the traditional forecast but has xwOBA −0.027, EV −2.8mph, and barrel −6.1pp; he is not a growth-signal priority. Shaw's wOBA +.025, xwOBA +.001, and barrel −2.8pp provide mixed support for the observed improvement.

APPENDIX B · LOW-HISTORY HITTERS

Retain visibility for hitters with insufficient prior history

Among 2026 hitters with at least 200PA, 63 did not meet the prior-season 100PA requirement. The 8 hitters below are modest-results candidates ordered by current xwOBA. This descriptive list receives no year-to-year change forecast and should not be assigned the same validation status as the forecast or experiment lists.

Hitter	PA	woba	xwoba	EV mph	Barrel%	Chase%	Whiff%
JJ Wetherholt	619	0.318	0.347	89.7	7.4	21.2	20.2
Spencer Jones	289	0.323	0.328	94.1	15.9	35.3	38.7
Ryan Vilade	278	0.327	0.327	89.2	7.1	27.4	23.6
Braden Montgomery	365	0.298	0.325	88.9	4.4	27.4	30.1
Jared Young	342	0.332	0.321	90.4	7.7	27.5	24.4
A.J. Ewing	492	0.311	0.320	89.6	7.5	26.7	24.6
Konnor Griffin	339	0.320	0.316	88.1	7.0	37.6	28.9
Cooper Pratt	266	0.320	0.315	87.0	3.2	29.8	21.8

APPENDIX C · EXPLORATORY EXTENSION DESIGN

Extensions to validate before adding more variables

Adding explanatory evidence and changing prediction models are separate tasks. This revision preserves M1–M3, all 298 forecasts, player IDs, and 2027 evaluation criteria. The extensions below have not been implemented and carry no performance-improvement claim.

Priority	Design	Evaluation conditions
1 · Multi-year baseline	Recent 3-year batting history and PA, sample-size regression to the mean, and age adjustment	Verify implementation against Marcel's public principles, then compare with B0 and M1 under identical time splits
2 · Candidate M4	M2 + prespecified multi-year summaries; park variables as a separate incremental comparison	Disclose incremental contributions, errors, and top-selection changes rather than adding all new groups together
3 · Environment and opportunity	Park, opposing-pitcher handedness, health, role, and playing-time opportunity	Use only information available by input-season end; model batting rate and PA availability separately
4 · Recent technical data	In-zone contact, sweet-spot%, batted-ball types, and bat tracking	Verify available years and denominators first; do not assume coverage across all of 2015–2026

Tune α only through time-ordered internal validation within training data preceding each outer evaluation year. A new model chosen after revisiting 2026 is exploratory M4 and cannot replace the frozen 2027 version. Actual M4 implementation requires separate dated hashes for data, variables, baseline, α rules, candidates, and evaluation code. This report alone does not register or validate M4.

Marcel uses recent 3-year history, weighting, regression to the mean, and age adjustment; its exact constants were not implemented here. Savant bat tracking covers the second half of 2023 onward [E2]; MLB introduced the public release on May 13, 2024 [E3]. Neither covers the entire 2015-2026 sample.

APPENDIX D · TERMS & NOTATION

Terms and success definitions

Term	Meaning in this report
PA	Plate appearances; distinct from at-bats (AB), the batting-average denominator.
wOBA	Batting metric weighting offensive outcomes by run contribution. [G1]
xwOBA	Expected wOBA using exit velocity, launch angle, sprint speed on some batted balls, and actual walks and strikeouts. [G2]
EV	Speed of a batted ball as it leaves the bat (mph).
Barrel%	Share of batted balls combining high exit velocity with an appropriate launch angle. The denominator in this dataset is batted balls. [G3]
Hard-hit%	Share of batted balls with exit velocity ≥95mph. [G4]
Chase%	Swings at pitches outside the strike zone as a share of outside-zone pitches. [G5]
Whiff%	Swinging misses as a share of all swings. [G6]
K%, BB%	Strikeouts and walks per plate appearance.
ISO	Slugging percentage−batting average; a measure of extra-base power. [G7]
Launch angle	Vertical angle of a batted ball at contact. The mean is distinct from the angle distribution or sweet-spot share. [G8]
pp	Percentage points. 30%→28% is −2pp, not −2%.
RMSE, MAE	Prediction-error metrics; lower is better. RMSE places more weight on large errors.
AUC	Discrimination that ranks successes above failures. 0.5 is random-level discrimination; 1 is perfect. It is not a success probability.
Ridge / α	Regression penalizing the sum of squared coefficients / penalty strength. Applied after feature scaling. [G9]

Bootstrap	Repeated sample resampling to assess uncertainty. This report’s AUC analysis resamples all records for each hitter together.
B0 / M1 / M2 / M3	Current-wOBA carry-forward / traditional results / added skill levels / added skill changes.
Breakout / retention	Input-season wOBA rise ≥.020 / next-season PA ≥200 and wOBA loss ≤.010.
Growth success / usable bat	Next-season PA ≥200 and wOBA rise ≥.020 / PA ≥200 and wOBA ≥.330. These differ from retention.
Investigation slot / screen	Budget unit for reviewing one player-season / rule filtering candidates by signals. Neither measures dollars or acquisition value.
Discovery pool	Base candidates defined by results, HR, and recent history. It does not directly measure attention or market undervaluation.

APPENDIX E · DECISION FOLLOW-UP

What decisions follow the small 2027 sample?

Even 4/5 or 5/5 successes in 2027 alone will not establish general superiority or replace M1. M1 remains the default, with at most 5 exploratory signal investigations. The rules below are retained from the supplied PDF. Its earlier addendum file and hash were absent from the supplied ZIP; no original registration claim is made.

Decision element	Rules specified before future outcomes
Cumulative review	First review: 2027–2029 (at most 15 slots). Accumulate growth successes, all selected slots, and 200PA availability by policy. If neither the pilot nor stopping conditions below apply, keep the same rules through 2031 (at most 25 slots).
Annual freeze	Same variables, α=10, screen, and outcome definitions. Refit M1 using only completed prior outcomes; preserve inputs, candidates, and selected IDs before each season. Comparators use the same annual k.
Limited pilot adoption	Screen growth rate ≥10pp above M1, random upper tail ≤.025, and PA availability no more than 10pp below M1. Retain a budget of 5 hitters.
Stop priority filtering	If growth rate or PA availability is ≥10pp below M1, stop using the screen as a priority filter and retain it only for individual diagnosis. This does not establish general inferiority.
Inconclusive results	Continue unchanged after 2029. If still inconclusive in 2031, retain M1 and do not adopt the priority filter. Do not extend until favorable results appear.

The .025 criterion at each of two reviews is a one-sided upper-tail probability P(random successes ≥ observed). The earlier benchmark table instead uses a two-sided probability-ordered tail; these are different quantities. No extra annual adoption tests apply. The 10pp margins are analyst judgments; these operating rules do not prove superiority over M1. Usable-bat yield cannot replace growth criteria.

Illustration: With independent binomial sampling, random rate .24, and n=5, P(≥3)=.0933 and P(≥4)=.0134. If the true rate is .40, power to reach 4 is 8.7%. At n=15 and n=25, the .025 upper-tail rule requires 8 and 11 successes; power is 21.3% and 41.4%, respectively. Adequate power is not guaranteed. The future benchmark will use each season’s finite pool and success count rather than fixing .24.

Retained rules: docs/DECISION_FOLLOWUP_RULES.md. Recomputed illustration: supplement/policy_power_illustration.csv. Current integrity checks: supplement/preservation_check.json. The original forecast and evaluation lock files are preserved.

Python implementation and reproducibility evidence

Python implementation and notebook outputs are included in the companion ZIP. No public GitHub URL is supplied. The example below uses the same time split and fitting pipeline as src/study.py, with separate DataFrame and column-list names.

```
feature_cols = MODELS["M1_results"]
train = features.loc[features.eligible_pair & features.target_season.lt(year)]
test = features.loc[features.eligible & features.target_season.eq(year)].copy()
assert train.target_season.max() < year
model = make_pipeline(
    SimpleImputer(strategy="median"), StandardScaler(), Ridge(alpha=ALPHA)
)
model.fit(train[feature_cols].to_numpy(dtype=float),
          train.target_woba.to_numpy(dtype=float))
```

Implementation to review	Path in ZIP
Models, time splits, and adoption rules	src/study.py
Sensitivity, AUC, and policy comparisons	src/robustness.py, src/revision.py
Source retrieval and metadata	src/download.py, data/raw/*.meta.json
Executed notebook	notebooks/Signals_for_Next_Season_2027.ipynb
Additional diagnostics and revised report	src/review_verification.py, src/review_presentation.py
Frozen evaluation, forecasts, and hashes	src/evaluate_2027.py, reports/2027_evaluation_lock.json
Random benchmarks, power, and retained follow-up rules	src/review_verification.py; docs/DECISION_FOLLOWUP_RULES.md

Code and notebook evidence do not establish university recognition of programming coursework. Frozen forecasts and evaluation hashes were checked against the supplied ZIP: supplement/preservation_check.json. Earlier supplementary source files were not supplied.

08 · LIMITS & REPRODUCIBILITY

What these results can establish

- Annual data do not test an advantage in within-season discovery timing or causal effects of behavior change.
- 2027 outcomes are future observations. All result fields remain pending; no realized 2027 hit rate is available.
- Forecasting batting rates among hitters reaching 200PA next season involves survival and opportunity selection. Policy backtests count insufficient PA as failure but do not directly predict playing time.
- 2020 was excluded as an input and target season. The shortened 2020 season may still be the prior comparison for 2021 changes; results excluding the affected 2022 test are disclosed in CSVs.
- Historical Statcast values retrieved today may have been recalculated. This is not a reconstruction of archived metrics available at each historical decision date.
- Repeated hitters and only 4 development years limit inference. Park, opposing-pitcher handedness, health, role, defense, and baserunning were not fully controlled.
- The discovery filter was refined after inspecting the first player list. The original 13-hitter and revised 5-hitter signal rules, additional sensitivity checks, and policy comparisons are exploratory, with no independent evaluation before 2027. Earlier changes are recorded in PROTOCOL.md and REVISION_LOG.md; this correction is documented in docs/REVIEW_CORRECTIONS.md.

Source checks

2026 Savant total PA of 183,849 matches the MLB season total. Differences of ±1PA remain for 8 hitters, with cause unresolved. Savant measurements are used consistently and discrepancies disclosed; none changes 200PA eligibility. Custom/Expected PA and wOBA agree. Some small-sample xwOBA rounding differences of 0.001 were reconciled to the Expected source. URLs, retrieval timestamps, and SHA256 hashes are retained for all 41 source requests.

Using the files

The original notebook reproduces the core analysis and Korean HTML. Run python -m src.review_verification to reproduce the new raw-data, α, random-benchmark, and power checks. Run python -m src.review_presentation to apply the PDF corrections (PyMuPDF required). The uploaded ZIP omitted earlier supplementary sources mentioned by the English PDF; this package does not claim to restore those original files. Current outputs and retained follow-up rules are included. Frozen forecasts, raw sources, and original evaluation locks are unchanged.

1. MLB Baseball Savant · Custom leaderboard — 2015–2026 regular season. Contact quality, swing metrics, and traditional results.
2. MLB Baseball Savant · Expected statistics — Canonical xwOBA source.
3. MLB · Expected wOBA — Metric definition. xwOBA summarizes observed contact quality and other inputs; it does not guarantee future performance.
4. MLB Stats API · 2026 hitting — Season PA reconciliation; people and schedule APIs also used.
5. [G1] MLB · wOBA · Definitions and extension-design references, accessed 2026-10-04.
6. [G2] MLB · xwOBA · Definitions and extension-design references, accessed 2026-10-04.
7. [G3] MLB · Barrel · Definitions and extension-design references, accessed 2026-10-04.
8. [G4] MLB · Hard-hit · Definitions and extension-design references, accessed 2026-10-04.
9. [G5] MLB · Chase definition · Definitions and extension-design references, accessed 2026-10-04.
10. [G6] MLB · Whiff definition · Definitions and extension-design references, accessed 2026-10-04.
11. [G7] MLB · ISO · Definitions and extension-design references, accessed 2026-10-04.
12. [G8] MLB · Launch angle · Definitions and extension-design references, accessed 2026-10-04.
13. [G9] scikit-learn · Ridge · Definitions and extension-design references, accessed 2026-10-04.
14. [E1] Tom Tango · Marcel · Definitions and extension-design references, accessed 2026-10-04.
15. [E2] Baseball Savant - Bat Tracking: availability from the second half of 2023. Accessed 2026-10-04.
16. SciPy · hypergeom / binom · Formulas for random-selection distributions and illustrative power.

[E3] MLB - Statcast bat tracking introduction, May 13, 2024.

[P1] Baseball Savant - Lars Nootbaar: 2026 team splits, PA, and wOBA.

[P2] Field Level Media / Reuters - Nootbaar injury recovery and June 5 return, August 3, 2026.

[P3] Arizona Diamondbacks - Nootbaar acquisition announcement, August 3, 2026.